

Distractor effects in decision making depends on individual's style of integrating choice attributes

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint posted

November 9, 2023 (this version)

Posted to bioRxiv

August 21, 2023

Sent for peer review

August 2, 2023

Jing Jun Wong, Alessandro Bongioanni, Matthew F S Rushworth, Bolton K H Chau 

Department of Rehabilitation Sciences, The Hong Kong Polytechnic University, Hong Kong • Cognitive

Neuroimaging Unit, CEA, INSERM, Université Paris-Saclay, NeuroSpin Center, 91191 Gif-sur-Yvette, France •

Department of Experimental Psychology, University of Oxford, Oxford OX1 3UD, UK • University Research Facility

in Behavioral and Systems Neuroscience, The Hong Kong Polytechnic University, Hong Kong

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

It is widely agreed that people make irrational decisions in the presence of irrelevant distractor options. However, there is little consensus on whether decision making is facilitated or impaired by the presence of a highly rewarding distractor or whether distraction effect operates at the level of options' component attributes rather than at the level of their overall value. To reconcile different claims, we argue that it is important to incorporate consideration of the diversity of people's ways of decision making. We focus on a recent debate over whether people combine choice attribute in an additive or multiplicative way. Employing a multi-laboratory dataset investigating the same decision making paradigm, we demonstrated that people used a mix of both approaches and the extent to which approach was used varied across individuals. Critically, we identified that this variability was correlated with the effect of distractor on decision making. Individuals who tended to use a multiplicative approach, and hence focused on overall value, showed a positive distractor effect. In contrast, in individuals who tended to use an additive approach, driven by component attributes, the opposite negative distractor effect (divisive normalisation) was prominent. These findings suggest that distractor effects can operate at the level of overall choice values and concur with recent behavioural and neuroscience findings that multiple distractor effects co-exist.

eLife assessment

This manuscript provides a **useful** demonstration that distractor effects in multi-attribute decision-making correlate with the form of attribute integration (additive vs. multiplicative). The evidence supporting the conclusions is generally **solid**, but there are concerns regarding the robustness of the statistical analyses. In addition, the lack of a clear theoretical motivation complicates the interpretation. The manuscript will be interesting to decision-making researchers in neuroscience, psychology, and related fields.

Introduction

Psychologists, economists and neuroscientists have been interested in whether and how decision making is influenced by the presence of unchooseable distractor options. Rationally, choices ought to be unaffected by distractors, however, it has been demonstrated repeatedly that this is not the case in human decision making. For example, the presence of a strongly rewarding, yet unchooseable, distractor can either facilitate or impair decision making (Louie et al., 2013 [↗](#); Chau et al., 2014 [↗](#); Webb et al., 2020 [↗](#)). Which effect predominates depends on the distractor's interactions with the chooseable options which, in turn, are a function of their values (Chau et al., 2020 [↗](#)).

Intriguingly, most investigations have considered the interaction between distractors and chooseable options either at the level of their overall utility or at the level of their component attributes, but not both. Recently, however, one study has considered both possible levels of option-distractor interactions and argued that the distractor effect operates mainly at the attribute level rather than the overall utility level (Cao and Tsetsos, 2022 [↗](#)). When the options are comprised of component attributes (for example one feature might indicate the probability of an outcome if the option is chosen while the other might indicate the magnitude of the outcome), it is argued that the distractor exerts its effects through interactions with the attributes of the chooseable options. However, as has been argued in other contexts, just because one type of distractor effect is present does not preclude another type from existing; different distractor effects are not mutually exclusive (Chau et al., 2020 [↗](#); Kohl et al., 2023 [↗](#)).

Moreover, the fact that people have diverse way of making decisions is often overlooked. We argue that this diversity can make different forms of distractor effect more or less prominent in different circumstances.

Multiple distractor effects have been proposed. At the level of the overall utility of the choice, the divisive normalization model suggests that the presence of a valuable distractor can impair decision accuracy and this is sometimes known as a negative distractor effect (Louie et al., 2013 [↗](#); Webb et al., 2020 [↗](#); Kohl et al., 2023 [↗](#)). Conversely, an attractor network model suggests the opposite – a valuable distractor can improve decision accuracy by slowing down decision speed and this is sometimes known as a positive distractor effect (Chau et al., 2014 [↗](#); Chang et al., 2019 [↗](#); Chau et al., 2020 [↗](#); Kohl et al., 2023 [↗](#)). At the other level, the level of choice attribute, the selective integration model emphasizes that individual attributes of the distractor interact with individual attributes of the chooseable options and distort the way they are integrated. Although these models are sometimes discussed as if they were mutually exclusive, it is possible that some, if not all, of them can co-exist. For example, in a two-attribute decision making scenario, each option and distractor fall at different points in a two-dimensional decision space defined by the two component attributes. Whether the distractor facilitates or impairs decision accuracy depends on the exact locations of the chooseable options and distractor (Dumbalska et al., 2020 [↗](#)). Alternatively, the decision space can be defined by the sum of and differences in the two chooseable options' overall values, and it has been shown that positive and negative distractor effects predominate in different parts of this decision space (Chau et al., 2020 [↗](#); Kohl et al., 2023 [↗](#)). Hence, it is unlikely that the distractor affects decision making in a single, monotonic way. In addition, although most studies focused on analyzing distractor effects at the group level, they often involved a mix of individuals showing positive and negative distractor effects (Chang et al., 2019 [↗](#); Webb et al., 2020 [↗](#)). It is possible that such variability in distractor effects could be related to variability in people's ways of combining attributes during decision making.

Options that we choose in everyday life are often multi-attribute in nature. For example, a job-seeker may consider the salary and the chance of successfully getting the job before submitting the application. An ideal way of combining the two attributes (the salary and success rate) is by calculating their product, i.e. employing the Expected Value model or a multiplicative rule (von Neumann and Morgenstern, 1944). However, it has been argued that, instead of following this ideal rule, people use a simpler additive rule to combine attributes, in which individual attributes of the same option are added together often via a weighted-sum procedure (Farashahi et al., 2019; Cao and Tsetsos, 2022). This is equivalent to assuming that people compare pairs of attributes separately, without integration. As such, it is easy to take a dichotomous view that people either use a multiplicative or additive rule in their decision making. Intriguingly, however, recently it has been shown that, at the level of each individual human or monkey, decision making involves a combination of both rules (Scholl et al., 2014; Bongioanni et al., 2021). The two computational strategies may rely on distinct neuronal mechanisms, one in parietal cortex estimating the perceptual differences between stimuli, leading to an additive rule, and a more sophisticated one in prefrontal cortex tracking integrated value, leading to a multiplicative rule. It is possible to devise a composite model by having a single parameter (an integration coefficient) to describe, for each individual decision maker, the extent to which decision-making is based on multiplication/addition. In other words, the integration coefficient captures individual differences in people's and animals' degree of attribute combination during decision making (Figure 1). Here we consider whether such individual differences could result in different forms of distractor effect.

In the current study, we re-analysed data collected from three different laboratories that involved 144 human participants choosing between two options in the presence of a distractor (Figure 1a and b) (Chau et al., 2014; Gluth et al., 2018; Chau et al., 2020). Recently, the data has been fitted using a multiplicative rule, additive rule, and a multiplicative rule with divisive normalization (Cao and Tsetsos, 2022). It was argued that participants' choice behaviour was best described by the additive rule and that the previously reported positive distractor effect was absent when utility was estimated using the additive rule. Here, we fitted the data using the same models and procedures, but also considered an additional composite model to capture individual variations in the relative use of multiplicative and additive rules (Figure 1c). We found that this compositive model provides the best account of participants' behaviour. Critically, those who employed a more multiplicative style of integrating choice attributes also showed stronger positive distractor effects, whereas those who employed a more additive style showed negative distractor effects. These findings concur with neural data demonstrating that the medial prefrontal cortex computes the overall values of choices in ways that go beyond simply adding their components together, and it is the neural site at which positive distractor effects emerge (Barron et al., 2013; Chau et al., 2014; Noonan et al., 2017; Papageorgiou et al., 2017; Fouragnan et al., 2019; Bongioanni et al., 2021).

Results

This study analysed empirical data acquired from human participants (five datasets; N=144) performing the multi-attribute decision-making task described in Figure 1 (Chau et al., 2014; Gluth et al., 2018).

Participants were tasked to maximise their rewards by choosing between options that were defined by different reward magnitudes (X) and probabilities (P) depicted as rectangular bars of different colours and orientations, respectively (Figure 1b). The task involved, as a control, Two-Option Trials, on which participants were offered two options. It also involved Distractor Trials, on which two chooseable options were presented alongside a distractor option which could

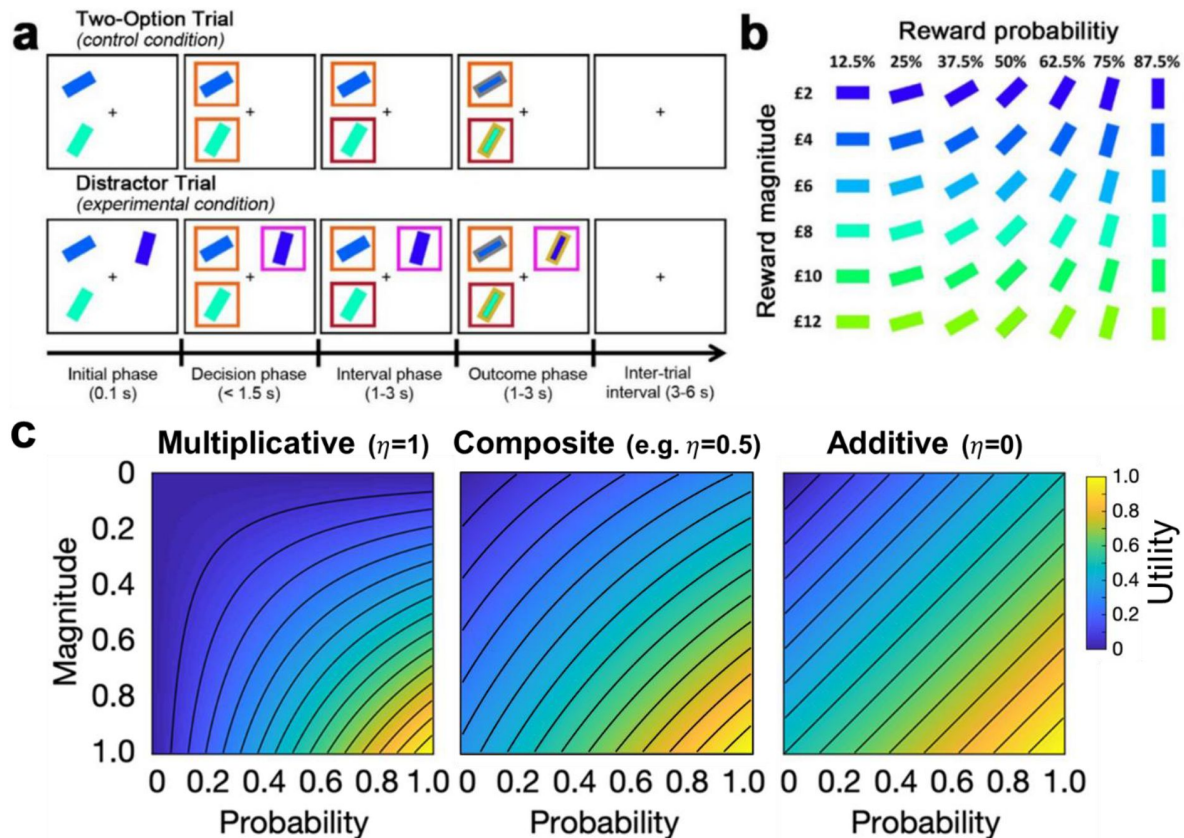


Figure 1.

The multi-attribute decision making task (Chau et al., 2014; Gluth et al., 2018). **(a)** On the control Two-Option Trials, participants were presented two options associated with different levels of reward magnitude and probability, each in the form of a rectangular bar surrounded by an orange box. On the experimental Distractor Trials, three options were presented. Two options were subsequently surrounded by orange boxes indicating that they could be chosen, while a third option was surrounded by a purple box indicating that it was an unchooseable distractor. **(b)** The association between stimulus colour and orientation to reward magnitude and probability, respectively. Participants were instructed to learn these associations prior to task performance. **(c)** Plots illustrating utility estimated using (left) a purely multiplicative rule and (right) a purely additive rule (here assuming equal weights for probability and magnitude). By comparing their corresponding plots, differences in the utility contours are most apparent in the bottom-left and top-right corners. This is because in the multiplicative rule a small value in either the magnitude or probability produces a small overall utility (blue colours). In contrast, in the additive rule the two attributes are independent – a small value in one attribute and a large value in another attribute can produce a moderate overall utility (green colours). (Middle) Here, we included a composite model that combines both rules. The composite model involves an integration coefficient η that controls the relative contributions of the multiplicative and additive rules.

Figure 1a and **b** are reproduced from (Chau et al., 2014), Nature Neuroscience.

not be selected. The two chooseable options and unchosen distractor option are referred to as the higher value (*HV*), lower value (*LV*), and distractor value (*DV*), based on their utility. We began our analyses by assuming utility as Expected Value (i.e., $EV = X \times P$).

A distractor effect was, on average, absent?

We used a general linear model (GLM), GLM1, to examine whether a distractor effect was present in the choice behaviour of participants. GLM1 included 3 regressors to predict the choice of the *HV* option: an *HV* – *LV* term that represents the choice difficulty (i.e., as the difference in value between *HV* and *LV* becomes smaller, it becomes more difficult to select the better option), a *DV* – *HV* term that represents the relative distractor value, and a $(HV - LV)(DV - HV)$ interaction term that examines whether the distractor effect was modulated as a function of choice difficulty. Similar approaches have been used previously (Chau et al., 2014 [↗](#); Gluth et al., 2018 [↗](#); Chau et al., 2020 [↗](#); Cao and Tsetsos, 2022 [↗](#); Kohl et al., 2023 [↗](#)). On Distractor Trials (**Figure 2a** [↗](#)), the results of GLM1 showed a positive *DV* – *HV* effect [$\beta = 0.0835$, $t(143) = 3.894$, $p = 0.000151$] and *HV* – *LV* effect [$\beta = 0.486$, $t(143) = 20.492$, $p < 10^{-43}$], and a negative $(HV - LV)(DV - HV)$ interaction effect [$\beta = -0.0720$, $t(143) = -3.668$, $p = 0.000344$]. However, Cao and Tsetsos (Cao and Tsetsos, 2022 [↗](#)) suggested that the same analysis should be applied to the control Two-Option Trials. As such, each control trial should be analysed by including a hypothetical distractor identical to the actual distractor that was present in the matched experimental Distractor Trials. Moreover, Cao and Tsetsos suggest specific methods for trial matching (Cao and Tsetsos, 2022 [↗](#)). Theoretically, a distractor effect should be absent when GLM1 is applied to analyse these control trials because the distractor was not really present. Surprisingly, although consistent with the finding of Cao and Tsetsos, the Two-Option Trials (**Figure 2b** [↗](#)) displayed a significant *DV* – *HV* effect [$\beta = 0.0685$, $t(143) = 2.877$, $p = 0.00463$] and a $(HV - LV)(DV - HV)$ interaction effect [$\beta = -0.0487$, $t(143) = -2.477$, $p = 0.0144$], alongside the expected *HV* – *LV* effect [$\beta = 0.635$, $t(143) = 27.975$, $p < 10^{-59}$].

One key test to examine whether a distractor effect was present was to compare the strength of distractor effects on the experimental Distractor Trials and the control Two-Option Trials. This was achieved by adapting GLM1 into GLM2 and then matching the control Two-Option Trials and Distractor Trials (here we followed exactly the approach suggested by Cao and Tsetsos. More details are presented in the section *GLM analysis of relative choice accuracy*). In addition to the regressors involved in GLM1, GLM2 included a binary variable *T* to describe the trial type (i.e., 0 = Two-Option Trials and 1 = Distractor Trial). Then all original GLM1 regressors were multiplied by this variable *T*. Hence, the presence of a “stronger” distractor effect on Distractor Trials than control trials should be manifested in the form of a significant $(DV - HV)T$ effect or $(HV - LV)(DV - HV)T$ effect. However, the results showed that neither the $(DV - HV)T$ nor $(HV - LV)(DV - HV)T$ term was significant. While these results may seem to suggest that a distractor effect was not present at an overall group level, we argue that the precise way in which a distractor affects decision making depends on how individuals integrate the attributes.

Identifying individual variabilities in combining choice attributes

During multi-attribute decision making, people integrate information from different attributes. However, the method of integration can be highly variable across individuals and this, in turn, has an impact on how a distractor effect manifests. Hence, it is imperative to first consider individual differences in how participants integrate the choice attributes. Previous work has demonstrated that choice behaviour may not be adequately described using either additive or multiplicative models, but rather a combination of both (Bongioanni et al., 2021 [↗](#)). We fitted these models to the Two-Option Trial data, such that the fitting was independent of any distractor effects, and tested which model best describes participants’ choice behaviours. In particular, we included the same set of models suggested by Cao and Tsetsos, which included an EV model, an additive utility (AU) model, and an EV model combined with divisive normalisation. In addition, we included a

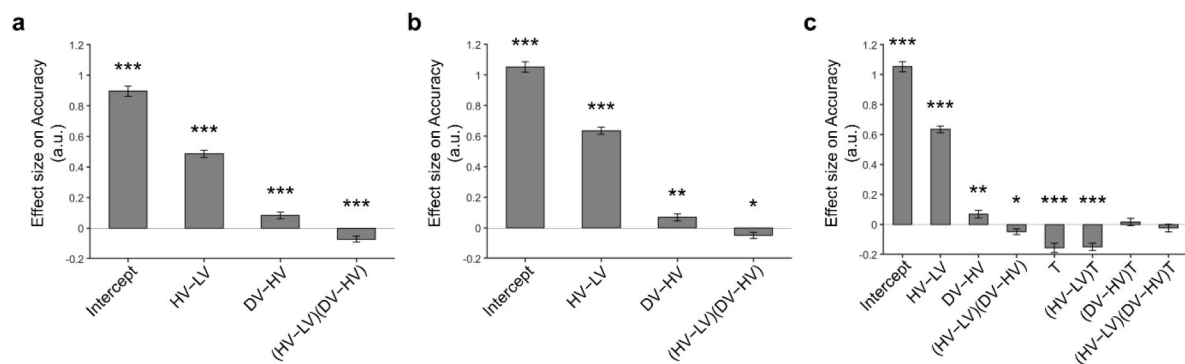


Figure 2.

A distractor effect was absent, when the way participants combined choice attributes was ignored. Although a positive distractor effect (indicated by the DV-HV term) was present on both **(a)** Distractor and **(b)** Two-Option Trials, **(c)** the distractor effects in the two trial types were not significantly different when they were compared in a single analysis [indicated by the $(DV - HV)T$ term]. Error bars indicate standard error. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

composite model which utilises an integration coefficient to incorporate both additive and multiplicative methods of combining option attributes. A larger integration coefficient suggests an individual places more weight on using a multiplicative rule than an additive rule.

When a Bayesian model comparison was performed, the results showed that the composite model provides the best account of participants' choice behaviour (**Figure 3c**; exceedance probability = 1.000, estimated model frequency = 0.8785). **Figures 3c** and **d** show the fitted parameters of the composite model: η , the integration coefficient determining the relative weighting of the additive and multiplicative value ($M = 0.324$, $SE = 0.0214$); γ , the magnitude/probability weighing ratio ($M = 0.415$, $SE = 0.0243$); and θ , the inverse temperature ($M = 9.643$, $SE = 0.400$). Our finding that the average integration coefficient η was 0.325 coincides with previous evidence that people were biased towards using an additive, rather than a multiplicative rule. However, it also shows rather than being fully additive ($\eta=0$) or multiplicative ($\eta=1$), people's choice behaviour is best described as a mixture of both.

Multiplicative style of integrating choice attributes was associated with a significant positive distractor effect

It has been shown that evaluations of choices driven by more than just an additive combination of attribute features depend on the medial and/or adjacent ventromedial prefrontal cortex (Bongioanni et al., 2021; Papageorgiou et al., 2017). On the other hand, a positive distractor effect is also linked to the modulation of decision signals in a similar prefrontal region (Chau et al., 2014). On the basis of these findings, we expected that a positive distractor effect may be highly related to the use of a multiplicative method of choice evaluation given their similar anatomical associations. As such, we proceeded to explore how the distractor effect (i.e., the effect of $(DV - HV)T$ obtained from GLM2; **Figure 2c**) was related to the integration coefficient (η) of the optimal model via a Spearman's rank correlation (**Figure 4**). As expected, a significant positive correlation was observed [$r(142) = 0.286$, $p = 0.000548$].

This correlation could be driven by three possible patterns of distractor effect. First, *greater* integration coefficients (i.e., being more multiplicative) could be related to more *positive* distractor effects. Second, smaller integration coefficients (i.e., being more additive) could be related to more *negative* distractor effects. Third, both positive and negative distractor effects could be present, but appeared separately in predominantly multiplicative and additive individuals respectively. To test which was the case, we used the mean integration coefficient value to divide the participants into two groups [Multiplicative Group ($N = 71$) and Additive Group ($N = 73$)]. We then analysed the data from each group using GLM2 (**Figures 5b** and **c**). We found that the distractor effect, reflected in the $(DV - HV)T$ term, was significantly greater in the Multiplicative Group than the Additive Group (**Figure 5**; $t(142) = 3.792$, $p = 0.00022$). Critically, the distractor effects were significant within each of the individual groups but bore opposite signs: the distractor effect was positive in the Multiplicative Group [$\beta = 0.105$, $t(70) = 3.438$, $p = 0.000991$] but negative in the Additive Group [$\beta = -0.0725$, $t(72) = -2.053$, $p = 0.0437$].

Finally, we performed two additional analyses that revealed comparable results to those shown in **Figure 5**. In the first analysis, reported in **Supplementary Figure 1**, we added an $HV + LV$ term to the GLM, because this term was included in some analyses of a previous study that used the same dataset (Chau et al., 2020). In the second analysis, reported in **Supplementary Figure 2**, we replaced the utility terms of GLM2. Since the above analyses involved using HV, LV, and DV values defined by the normative Expected Value model, here, we re-defined the values using the composite model prior to applying GLM2. The results of both analyses remained broadly comparable. In the Multiplicative Group a significant positive distractor effect was found in both analyses. In the Additive Group a significant negative distractor effect was found in one analysis and a similar trend, despite not reaching significance, was found in another analysis.

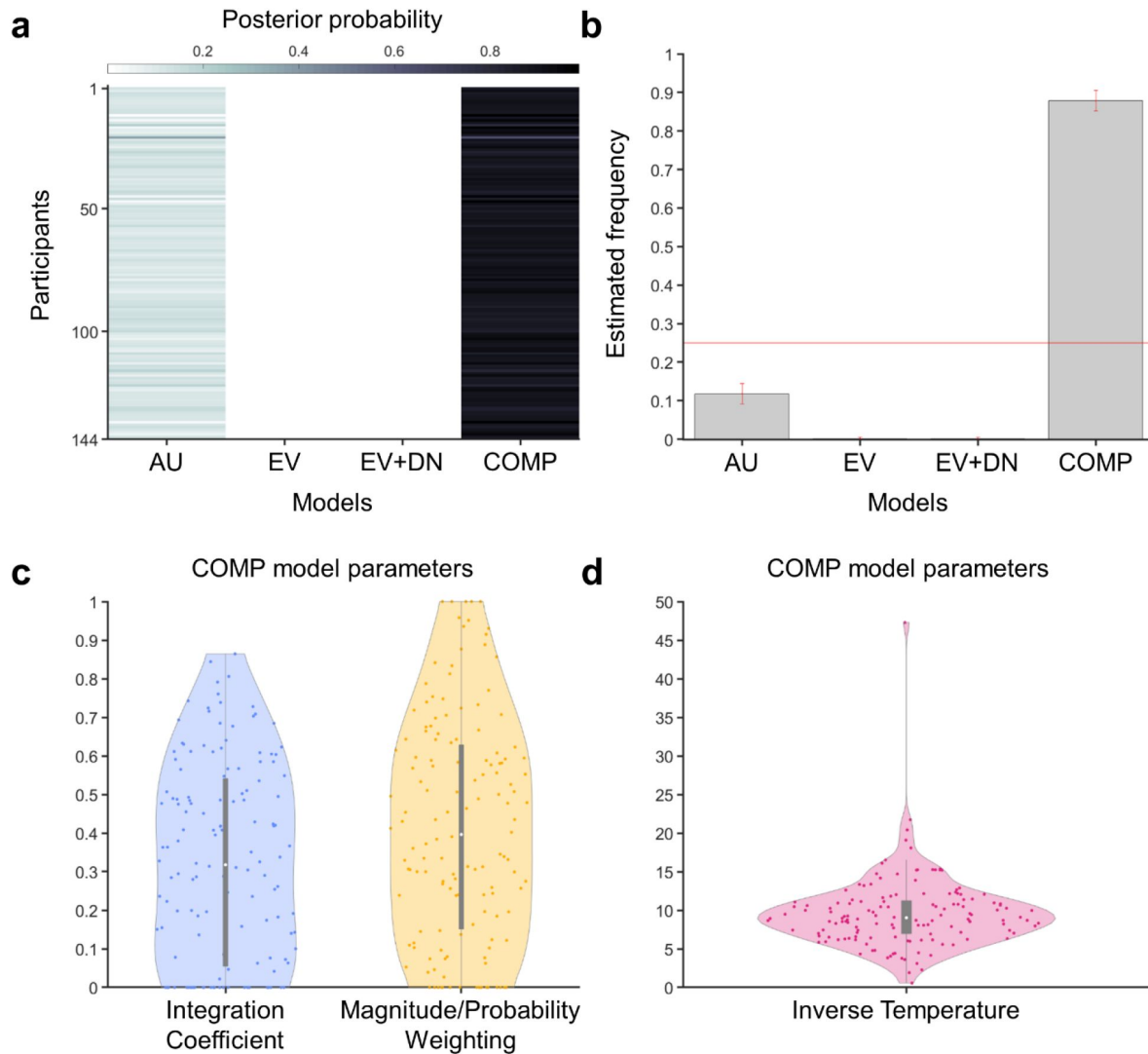


Figure 3.

A Bayesian model comparison showed that the composite model provides the best account of participants' choice behaviour. **(a)** The posterior probability of each model in accounting for the behaviour of each participant's behaviour. **(b)** A model comparison demonstrating that the composite model (COMP) is the best fit model, compared to the additive utility (AU), expected value (EV), and expected value and divisive normalisation (EV+DN) models. Violin plots showing each participant's fitted parameters using the composite model: **(c)** Values of the integration coefficient ($0 \leq \eta \leq 1$), magnitude/probability weighing ratio ($0 \leq \gamma \leq 1$), and **(d)** inverse temperature (θ).

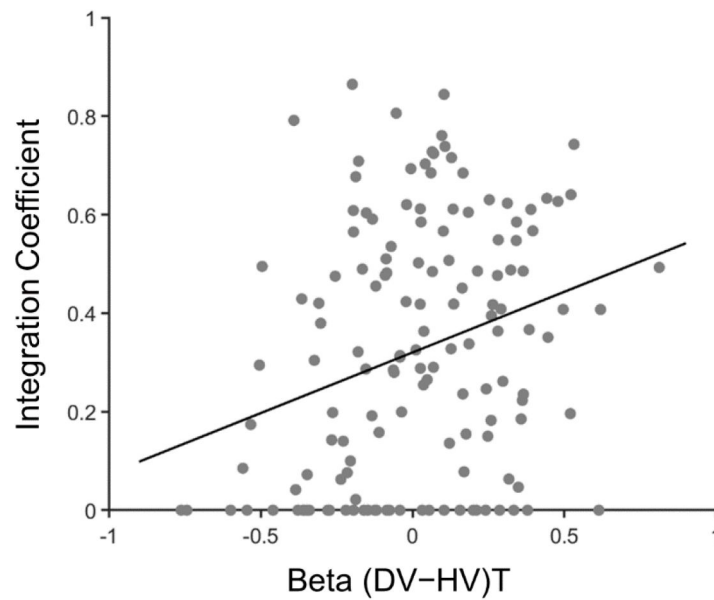


Figure 4.

Participants showing a more multiplicative style of combining choice attributes into an overall value also showed a greater distractor effect. A scatterplot illustrating a positive correlation between the integration coefficient (η) and the distractor term (i.e., $(DV - HV)T$) obtained from GLM2 (Figure 2c).

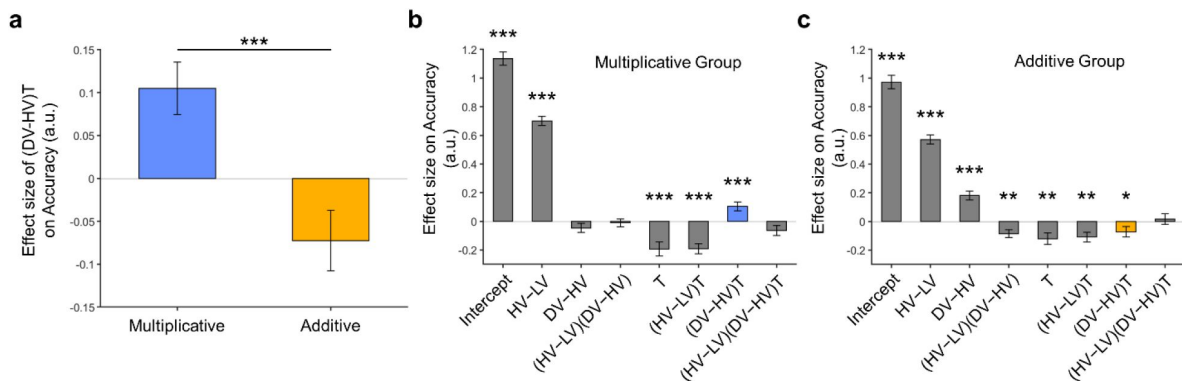


Figure 5.

Significant distractor effects were found when participants' styles of combining choice attributes were considered. **(a)** A positive distractor effect, indicated by the $(DV - HV)T$ term, was found in the Multiplicative Group, whereas a negative distractor effect was found in the Additive Group. The $(DV - HV)T$ term was also significantly different between the two groups. Plots showing all regression weights of GLM2 when the data of the **(b)** Multiplicative Group and **(c)** Additive Group were analysed. The key $(DV - HV)T$ term extracted in (a) is highlighted in blue and yellow for the Multiplicative Group and Additive Group, respectively. Error bars indicate standard error. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

It has been widely agreed that humans make irrational decisions in the presence of distractor options. However, there has been little consensus on how exactly distractors influence decision making. Does a seemingly attractive distractor impair or facilitate decision making? Does the distractor influence decision making at the level of overall utility or individual choice attributes? Often, one assumption behind these questions is that there is only a single type of distractor effect. Here, we demonstrated that, instead of a unidirectional effect, the way a distractor influences decision making depends on individual persons' styles of integrating choice attributes during decision making. More specifically, those who employed a multiplicative style of integrating attributes were prone to a positive distractor effect. Conversely, those who employed an additive style of integrating attributes were prone to a negative distractor effect. These findings show that the precise way in which distractors affect decision making depends on an interaction between the distractor value and people's style of decision making.

At the neuroanatomical level, the negative distractor effect is mediated by the posterior parietal cortex (PPC) (Louie et al., 2011 [DOI](#); Chau et al., 2014 [DOI](#)), but the same region is also crucial for perceptual decision making processes (Shadlen and Shohamy, 2016 [DOI](#)). The additive heuristics for combining choice attributes are closer to a perceptual evaluation because distances in this subjective value space correspond linearly to differences in physical attributes of the stimuli, whereas objective (multiplicative) value has a non-linear relation with them (cf. **Figure 1c** [DOI](#)). It is well understood that many sensory mechanisms, such as in primates' visual systems or fruitflies' olfactory systems, are subject to divisive normalization (Carandini and Heeger, 2012 [DOI](#)). Hence, the additive heuristics that are more closely based on sensory mechanisms could also be subject to divisive normalization, leading to negative distractor effects in decision making.

In contrast, the positive distractor effect is mediated by the medial prefrontal cortex (mPFC) (Chau et al., 2014 [DOI](#); Fouragnan et al., 2019 [DOI](#)), which is also crucial for representing abstract cognitive space. In situations that require combining multiple dimensions non-linearly, the mPFC integrates dimensions to achieve a closer approximation of objective value, sometimes using a "grid-like" code, such as during multi-attribute decision making, understanding social relations, and abstract knowledge (Constantinescu et al., 2016 [DOI](#); Bongioanni et al., 2021 [DOI](#); Park et al., 2021 [DOI](#)). Indeed, disrupting the mPFC in macaques also impaired their use of a multiplicative strategy. Hence, a positive distractor effect appears only when mPFC constructs the actual expected value non-linearly from the component attributes, a more refined strategy than the one followed by PPC.

Other studies have provided evidence suggesting that multiple forms of distractor effect can co-exist. For example, when people were asked to choose between three food items, most people were poorer at choosing the best option when the worst option (i.e. the distractor) was still an appealing option (Louie et al., 2013 [DOI](#); Webb et al., 2020 [DOI](#)). In other words, most people showed negative distractor effects, which is predicted by divisive normalization models (Carandini and Heeger, 2012 [DOI](#); Louie et al., 2013 [DOI](#)).

However, it is noticeable that the degree of negative distractor effect varies across individuals and some even showed the reverse, positive distractor effect. Similarly, it has been shown in social decision making, choice utility was best described by a divisive normalization model (Chang et al., 2019 [DOI](#)).

However, at the same time, a positive distractor effect was found in choice behaviour, such that greater distractor values were instead associated with more choices of the best option, suggesting a positive distractor effect. The effect became even more robust when individual variability was considered in a mixed effects model. Together, these results suggest that divisive normalization (which predicts negative distractor effects) and positive distractor effect may co-exist, but that they predominate in different aspects of decision making.

Indeed, it is possible that multiple forms of distractor effect can co-exist because of their different neuroanatomical origins. In human and monkey, a positive distractor effect was found in decision signals in the mPFC (Chau et al., 2014 [↗](#); Fouragnan et al., 2019 [↗](#)), whereas divisive normalization was found in decision signals in the PPC (Louie et al., 2011 [↗](#); Chau et al., 2014 [↗](#)). As such, it should be expected that while these opposite distractor effects might sometimes diminish one another, disruption in one of the brain regions might result in the expression of the distractor effect related to the other brain region.

Indeed, this idea is supported by empirical data that the parietal-related, negative distractor effect was more prominent in humans and monkeys with a lesion in the medial prefrontal cortex (Noonan et al., 2010 [↗](#); Noonan et al., 2017 [↗](#)). In addition, the prefrontal-related, positive distractor effect was more prominent after the parietal cortex was transiently disrupted using transcranial magnetic stimulation (Kohl et al., 2023 [↗](#)). These findings concur with the general notion that decision making is mediated by a distributed neural circuit, rather than a single, localized brain region.

Methods

Multi-attribute decision-making task and datasets

The current study re-analysed five published datasets of empirical data based on a multi-attribute decision-making task (Chau et al., 2014 [↗](#); Gluth et al., 2018 [↗](#)). The experimental task involved participants making a decision between two options in the absence (Two-Option Trials) or presence (Distractor Trials) of a third distractor option that could not be chosen. During each trial, two or three stimuli associated with different reward magnitudes and probabilities (represented by colours and orientations) were randomly presented in selected screen quadrants (**Figure 1** [↗](#)). Immediately following stimulus onset (0.1 s), options that were available for selection were surrounded by orange boxes, while the distractor option that could not be selected was surrounded by a purple box. The choice of the participant was indicated by the change in colour of the surrounding box from orange to red. At the end of each trial, the edge of each stimulus changed to yellow if the choice was rewarded, and to grey if the choice was not rewarded. A total of 144 human participants were included in the analysis with data from the original fMRI dataset ($N = 21$; (Chau et al., 2014 [↗](#))) and additional replication experiments (Experiments 1, 2, 3, and 4) performed by Gluth and colleagues ($N = 123$; (Gluth et al., 2018 [↗](#))).

GLM analysis of relative choice accuracy

The choice accuracy data from the Two-Option Trials and Distractor Trials were analysed separately using GLM1:

$$\text{Logit}(\text{accuracy}) = \beta_0 + \beta_1 z_{(HV-LV)} + \beta_2 z_{(DV-HV)} + \beta_3 z_{(HV-LV)} z_{(DV-HV)} + \varepsilon$$

where *HV*, *LV*, and *DV* refer to the values of the chooseable higher value option, chooseable lower value option, and distractor, respectively. $z_{(x)}$ refer to the z-scoring of term *x*, which were applied to all terms within the GLMs. The unexplained error is represented by ε . Trials in which the distractor or an empty quadrant were mistakenly chosen was excluded from the analysis. Fitting was performed using MATLAB's `glmfit` function by assuming a binomial distribution.

A binary variable (*T*) encoding trial type (i.e., 0 = Two-Option Trials and 1 = Distractor Trials) was introduced to combine both two-option and distractor trial types into a single GLM used to assess distractor-related interactions:

The choice accuracy data of the Two-Option Trials and Distractor Trials were analysed together using GLM2, which followed exactly the procedures described by Cao and Tsetsos (Cao & Tsetsos, 2022 [DOI](#)):

$$\begin{aligned} \text{Logit}(\text{accuracy}) &= \beta_0 + \beta_1 z_{(HV-LV)} + \beta_2 z_{(DV-HV)} + \beta_3 z_{(HV-LV)} z_{(DV-HV)} + \beta_4 T + \beta_5 z_{(HV-LV)} T \\ &+ \beta_6 z_{(DV-HV)} T + \beta_7 z_{(HV-LV)} z_{(DV-HV)} T + \varepsilon \end{aligned}$$

We focused on the relative accuracy of Distractor Trials compared to two choice trials by considering the proportion of H choices among trials with H or L choices and excluding a small number of D-choice trials. We also identified matched Two-Option Trials for each distractor trial. The 150 Distractor Trials yielded 149 unique conditions exhibiting a unique combination of probability (P) and magnitude (X) attributes across the three options (H, L, and D). However, the 150 Two-Option Trials contained only 95 unique conditions carrying a unique combination of P and X across the two available options (H and L). As the distractor and Two-Option Trials do not exhibit ‘one-to-one’ mapping, some Distractor Trials will have more than one matched Two-Option Trials. The different counts of matched Two-Option Trials were used as ‘observation weights’ in the GLM. Through identifying all matched Two-Option Trials of every distractor trial, the baselining approach introduces ‘trial-by-trial baseline accuracy’ as a new dependent variable as described by Cao and Tsetsos (Cao & Tsetsos, 2022 [DOI](#)).

Computational modelling

To determine which model best describes participants’ choice behaviour, we followed the procedures carried out by Cao and Tsetsos (Cao & Tsetsos, 2022 [DOI](#)). We fitted three models to choice data of the control, Two-Option Trials, namely the Expected Value (EV) model, Additive Utility (AU) model, and the Expected Value and Divisive Normalisation (EV+DN) model. In this study, we also included an additional composite model. For each model, we applied the softmax function as the basis for estimating choice probability:

$$p(\text{choice} = i) = \frac{e^{\vartheta U_i}}{\sum_{j=1}^n e^{\vartheta U_j}}$$

where U_i refers to the utility of option i (either the HV or LV option) and ϑ refers to the inverse temperature parameter. Four models were used to estimate the options’ utility U_i , based on their corresponding reward magnitude X_i and probability P_i (both rescaled to the interval between 0 and 1). The four models were the Expected Value model, Additive Utility model, Expected Value and Divisive normalisation model, and Composite model:

Expected Value (EV) model

This model employs a multiplicative rule for estimating utility:

$$U_i = X_i \times P_i$$

Additive Utility (AU) model

This model employs an additive rule for estimating utility based on the magnitude X_i and probability P_i as follows:

$$U_i = \gamma X_i + (1 - \gamma) P_i$$

where γ is the magnitude/probability weighing ratio ($0 \leq \gamma \leq 1$).

Expected Value and Divisive Normalisation (EV+DN) model

We included the EV+DN model following the procedures carried out by Cao and Tsetsos (Cao & Tsetsos, 2022 [↗](#)). Compared to the EV models, here utilities were normalised by the inputs of all values as follows:

$$EV_i = X_i \times P_i$$

$$U_i = \frac{EV_i}{\sum_{j=1}^N EV_j}$$

where EV_i , X_i , and P_i denotes expected value, reward magnitude, and reward probability of option i , respectively.

Composite model

We further explored the possibility that behaviour may be described as a mixture of both additive AU and multiplicative EV models (Scholl et al., 2014 [↗](#); Bongioanni et al., 2021 [↗](#)):

$$U_i = \eta(X_i \times P_i) + (1 - \eta)(\gamma X_i + (1 - \gamma)P_i)$$

where γ is the magnitude/probability weighing ratio ($0 \leq \gamma \leq 1$) and η is the integration coefficient ($0 \leq \eta \leq 1$) determining the relative weighting of the additive and multiplicative value. Simulations reported in Bongioanni et al., 2021 [↗](#) prove that the composite model can be accurately discriminated from the simple additive and multiplicative models, even if the latter ones include non-linear distortions of the magnitude and probability dimensions. Additionally, it was shown that the parameter estimate for the composite model is very accurate.

Model fitting and comparison

The empirical choice probabilities generated from the model predictions were used to calculate the binomial log-likelihood, following the procedures described by Cao and Tsetsos (Cao and Tsetsos, 2022 [↗](#)):

$$LL \propto p_e \log(p_m) + (1 - p_e) \log(1 - p_m)$$

Here, p_e and p_m represent the empirical and model-predicted $p(H \text{ over } L)$, respectively.

Model fitting was performed on the behavioural data to maximise the log-likelihood summed over trials. A grid of randomly generated starting values for free parameters of each model was used to fit each participant's data at least 10 times to avoid local optima. Model fitting was performed using MATLAB's `fmincon` function with the maximum number of function evaluations and iterations set to 5000 and the optimality and step tolerance set to 10^{-10} . The variational Bayesian analysis (VBA) toolbox (Daunizeau et al., 2014; Rigoux et al., 2014) was used to calculate each model's posterior frequency and protected exceedance probability. Aligning with the methods used by Cao and Tsetsos (Cao & Tsetsos, 2022 [↗](#)), only the trials where H or L responses were made were included in the analysis. Trials in which participants opted for the unchooseable D were excluded.

Five-fold cross-validation was performed for model comparison. First, this involved dividing the 150 trials for each participant into five folds. Four random folds of trials would be randomly selected to generate best-fitting parameters used to calculate the log-likelihood summed across trials in the unchosen fold. We repeated this process five times for each unchosen fold and computed the average log-likelihood across cross-validation folds to generate the cross-validated log-likelihood for each model. We then simulated each model's behaviour using the best-fitting parameters obtained during model fitting. The simulated behaviour was cross-fitted to all models to calculate the log-likelihoods summed over trials. Lastly, the goodness-of-fit for the models were evaluated using Bayesian model comparison.

Acknowledgements

This work was supported by the Hong Kong Research Grants Council (15105522) and Wellcome Trust grant 221794/Z/20/Z.

References

- Barron HC, Dolan RJ, Behrens TE (2013) **Online evaluation of novel choices by simultaneous representation of multiple memories** *Nat Neurosci* **16**:1492–1498
- Bongioanni A, Folloni D, Verhagen L, Sallet J, Klein-Flugge MC, Rushworth MFS (2021) **Activation and disruption of a neural mechanism for novel choice in monkeys** *Nature* **591**:270–274
- Cao Y, Tsetsos K (2022) **Clarifying the role of an unavailable distractor in human multiattribute choice** *eLife* **11**
- Carandini M, Heeger DJ (2012) **Normalization as a canonical neural computation** *Nat Rev Neurosci* **13**:51–62
- Chang LW, Gershman SJ, Cikara M (2019) **Comparing value coding models of context-dependence in social choice** *Journal of Experimental Social Psychology* **85**
- Chau BK, Kolling N, Hunt LT, Walton ME, Rushworth MF (2014) **A neural mechanism underlying failure of optimal choice with multiple alternatives** *Nat Neurosci* **17**:463–470
- Chau BK, Law CK, Lopez-Persem A, Klein-Flugge MC, Rushworth MF (2020) **Consistent patterns of distractor effects during decision making** *eLife* **9**
- Constantinescu AO, O'Reilly JX, Behrens TEJ (2016) **Organizing conceptual knowledge in humans with a gridlike code** *Science* **352**:1464–1468
- Dumbalska T, Li V, Tsetsos K, Summerfield C (2020) **A map of decoy influence in human multialternative choice** *Proc Natl Acad Sci U S A* **117**:25169–25178
- Farashahi S, Donahue CH, Hayden BY, Lee D, Soltani A (2019) **Flexible combination of reward information across primates** *Nat Hum Behav* **3**:1215–1224
- Fouragnan EF *et al.* (2019) **The macaque anterior cingulate cortex translates counterfactual choice value into actual behavioral change** *Nat Neurosci* **22**:797–808
- Gluth S, Spektor MS, Rieskamp J (2018) **Value-based attentional capture affects multi-alternative decision making** *eLife* **7**
- Kohl C, Wong MXM, Wong JJ, Rushworth MFS, Chau BKH (2023) **Intraparietal stimulation disrupts negative distractor effects in human multi-alternative decision-making** *eLife* **12**
- Louie K, Gratton LE, Glimcher PW (2011) **Reward value-based gain control: divisive normalization in parietal cortex** *J Neurosci* **31**:10627–10639
- Louie K, Khaw MW, Glimcher PW (2013) **Normalization is a general neural mechanism for context-dependent decision making** *Proceedings of the National Academy of Sciences*
- Noonan MP, Chau BKH, Rushworth MFS, Fellows LK (2017) **Contrasting Effects of Medial and Lateral Orbitofrontal Cortex Lesions on Credit Assignment and Decision-Making in Humans** *J Neurosci* **37**:7023–7035

Noonan MP, Walton ME, Behrens TE, Sallet J, Buckley MJ, Rushworth MF (2010) **Separate value comparison and learning mechanisms in macaque medial and lateral orbitofrontal cortex** *Proc Natl Acad Sci U S A* **107**:20547–20552

Papageorgiou GK, Sallet J, Wittmann MK, Chau BKH, Schuffelgen U, Buckley MJ, Rushworth MFS (2017) **Inverted activity patterns in ventromedial prefrontal cortex during value-guided decision-making in a less-is-more task** *Nature communications* **8**

Park SA, Miller DS, Boorman ED (2021) **Inferences on a multidimensional social hierarchy use a grid-like code** *Nat Neurosci* **24**:1292–1301

Scholl J, Gunthner J, Kolling N, Favaron E, Rushworth MF, Harmer CJ, Reinecke A (2014) **A role beyond learning for NMDA receptors in reward-based decision-making—a pharmacological study using d-cycloserine** *Neuropsychopharmacology* **39**:2900–2909

Shadlen MN, Shohamy D (2016) **Decision Making and Sequential Sampling from Memory** *Neuron* **90**:927–939

von Neumann J, Morgenstern O (1944) **Theory of Games and Economic Behavior**

Webb R, Glimcher PW, Louie K (2020) **Divisive normalization does influence decisions with multiple alternatives** *Nat Hum Behav* **4**:1118–1120

Article and author information

Jing Jun Wong

Department of Rehabilitation Sciences, The Hong Kong Polytechnic University, Hong Kong
ORCID iD: [0000-0001-5241-6997](https://orcid.org/0000-0001-5241-6997)

Alessandro Bongioanni

Cognitive Neuroimaging Unit, CEA, INSERM, Université Paris-Saclay, NeuroSpin Center, 91191 Gif-sur-Yvette, France

Matthew F S Rushworth

Department of Experimental Psychology, University of Oxford, Oxford OX1 3UD, UK

Bolton K H Chau

Department of Rehabilitation Sciences, The Hong Kong Polytechnic University, Hong Kong, University Research Facility in Behavioral and Systems Neuroscience, The Hong Kong Polytechnic University, Hong Kong

For correspondence: boltonchau@gmail.com

ORCID iD: [0000-0002-6854-5176](https://orcid.org/0000-0002-6854-5176)

Copyright

© 2023, Wong et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Thorsten Kahnt

National Institute on Drug Abuse Intramural Research Program, United States of America

Senior Editor

Joshua Gold

University of Pennsylvania, United States of America

Reviewer #1 (Public Review):

Summary:

The current study provided a follow-up analysis using published datasets focused on the individual variability of both the distraction effect (size and direction) and the attribute integration style, as well as the association between the two. The authors tried to answer the question of whether the multiplicative attribute integration style concurs with a more pronounced and positively oriented distraction effect.

Strengths:

The analysis extensively examined the impacts of various factors on decision accuracy, with a particular focus on using two-option trials as control trials, following the approach established by Cao & Tsetsos (2022). The statistical significance results were clearly reported.

The authors meticulously conducted supplementary examinations, incorporating the additional term HV+LV into GLM3. Furthermore, they replaced the utility function from the expected value model with values from the composite model.

Weaknesses:

There are several weaknesses in terms of theoretical arguments and statistical analyses.

First, the manuscript suggests in the abstract and at the beginning of the introduction that the study reconciled the "different claims" about "whether distraction effect operates at the level of options' component attributes rather than at the level of their overall value" (see line 13-14), but the analysis conducted was not for that purpose. Integrating choice attributes in either an additive or multiplicative way only reflects individual differences in combining attributes into the overall value. The authors seemed to assume that the multiplicative way generated the overall value ("Individuals who tended to use a multiplicative approach, and hence focused on overall value", line 20-21), but such implicit assumption is at odds with the statement in line 77-79 that people may use a simpler additive rule to combine attributes, which means overall value can come from the additive rule.

The second weakness is sort of related but is more about the lack of coherent conceptual understanding of the "additive rule", or "distractor effect operates at the attribute level". In an assertive tone (lines 77-80), the manuscript suggests that a weighted sum integration procedure of implementing an "additive rule" is equal to assuming that people compare pairs of attributes separately, without integration. But they are mechanistically distinct. The additive rule (implemented using the weighted sum rule to combine probability and magnitude within each option and then applying the softmax function) assumes value exists before comparing options. In contrast, if people compare pairs of attributes separately, preference forms based on the within-attribute comparisons. Mathematically these two might be equivalent only if no extra mechanisms (such as inhibition, fluctuating attention, evidence accumulation, etc) are included in the within-attribute comparison process, which is hardly true in the three-option decision.

Could the authors comment on the generalizability of the current result? The reward magnitude and probability information are displayed using rectangular bars of different colors and orientations. Would that bias subjects to choose an additive rule instead of the multiplicative rule? Also, could the conclusion be extended to other decision contexts such as quality and price, whether a multiplicative rule is hard to formulate?

The authors did careful analyses on quantifying the "distractor effect". While I fully agree that it is important to use the matched two-option trials and examine the interaction terms (DV-HV)T as a control, the interpretation of the results becomes tricky when looking at the effects in each trial type. Figure 2c shows a positive DV-HV effect in two-option trials whereas the DV-HV effect was not significantly stronger in three-option trials. Further in Figure 5b,c, in the Multiplicative group, the effect of DV-HV was absent in the two-option trials and present in the three-option trials. In the Additive group, however, the effect of DV-HV was significantly positive in the two-option trials but was significantly lowered in the three-option trials. Hence, it seems the different distractor effects were driven by the different effects of DV-HV in the two-option trials, rather than the three-option trials?

Note that the pattern described above was different in Supplementary Figure 2, where the effect of DV-HV on the two-option trials was negative for both Multiplicative and Additive groups. I would suggest considering using Supplementary Figure 2 as the main result instead of Figure 5, as it does not rely on multiplicative EV to measure the distraction effect, and it shows the same direction of DV-HV effect on two-option trials, providing a better basis to interpret the (DV-HV)T effect.

Reviewer #2 (Public Review):

This paper addresses the empirical demonstration of "distractor effects" in multi-attribute decision-making. It continues a debate in the literature on the presence (or not) of these effects, which domains they arise in, and their heterogeneity across subjects. The domain of the study is a particular type of multi-attribute decision-making: choices over risky lotteries. The paper reports a re-analysis of lottery data from multiple experiments run previously by the authors and other laboratories involved in the debate.

Methodologically, the analysis assumes a number of simple forms for how attributes are aggregated (adaptively, multiplicatively, or both) and then applies a "reduced form" logistic regression to the choices with a number of interaction terms intended to control for various features of the choice set. One of these interactions, modulated by ternary/binary treatment, is interpreted as a "distractor effect."

The claimed contribution of the re-analysis is to demonstrate a correlation in the strength/sign of this treatment effect with another estimated parameter: the relative mixture of additive/multiplicative preferences.

Major Issues

1. How to Interpret GLM 1 and 2

This paper, and others before it, have used a binary logistic regression with a number of interaction terms to attempt to control for various features of the choice set and how they influence choice. It is important to recognize that this modelling approach is not derived from a theoretical claim about the form of the computational model that guides decision-making in this task, nor an explicit test for a distractor effect. This can be seen most clearly in the equations after line 321 and its corresponding log-likelihood after 354, which contain no parameter or test for "distractor effects". Rather the computational model assumes a binary choice probability and then shoehorns the test for distractor effects via a binary/ternary

treatment interaction in a separate regression (GLM 1 and 2). This approach has already led to multiple misinterpretations in the literature (see Cao & Tsetsos, 2022; Webb et al., 2020). One of these misinterpretations occurred in the datasets the authors studied, in which the lottery stimuli contained a confound with the interaction that Chau et al., (2014) were interpreting as a distractor effect (GLM 1). Cao & Tsetsos (2022) demonstrated that the interaction was significant in binary choice data from the study, therefore it can not be caused by a third alternative. This paper attempts to address this issue with a further interaction with the binary/ternary treatment (GLM 2). Therefore the difference in the interaction across the two conditions is claimed to now be the distractor effect. The validity of this claim brings us to what exactly is meant by a "distractor effect."

The paper begins by noting that "Rationally, choices ought to be unaffected by distractors" (line 33). This is not true. There are many normative models that allow for the value of alternatives (even low-valued "distractors") to influence choices, including a simple random utility model. Since Luce (1959), it has been known that the axiom of "Independence of Irrelevant Alternatives" (that the probability ratio between any two alternatives does not depend on a third) is an extremely strong axiom, and only a sufficiency axiom for a random utility representation (Block and Marschak, 1959). It is not a necessary condition of a utility representation, and if this is our definition of rational (which is highly debatable), not necessary for it either. Countless empirical studies have demonstrated that IIA is falsified, and a large number of models can address it, including a simple random utility model with independent normal errors (i.e. a multivariate Probit model). In fact, it is only the multinomial Logit model that imposes IIA. It is also why so much attention is paid to the asymmetric dominance effect, which is a violation of a necessary condition for random utility (the Regularity axiom).

So what do the authors even mean by a "distractor effect." It is true that the form of IIA violations (i.e. their path through the probability simplex as the low-option varies) tells us something about the computational model underlying choice (after all, different models will predict different patterns). However we do not know how the interaction terms in the binary logit regression relate to the pattern of the violations because there is no formal theory that relates them. Any test for relative value coding is a joint test of the computational model and the form of the stochastic component (Webb et al, 2020). These interaction terms may simply be picking up substitution patterns that can be easily reconciled with some form of random utility. While we can not check all forms of random utility in these datasets (because the class of such models is large), this paper doesn't even rule any of these models out.

1. How to Interpret the Composite (Mixture) model?

On the other side of the correlation are the results from the mixture model for how decision-makers aggregate attributes. The authors report that most subjects are best represented by a mixture of additive and multiplicative aggregation models. The authors justify this with the proposal that these values are computed in different brain regions and then aggregated (which is reasonable, though raises the question of "where" if not the mPFC). However, an equally reasonable interpretation is that the improved fit of the mixture model simply reflects a misspecification of two extreme aggregation processes (additive and EV), so the log-likelihood is maximized at some point in between them.

One possibility is a model with utility curvature. How much of this result is just due to curvature in valuation? There are many reasonable theories for why we should expect curvature in utility for human subjects (for example, limited perception: Robson, 2001, Khaw, Li Woodford, 2019; Netzer et al., 2022) and of course many empirical demonstrations of risk aversion for small stakes lotteries. The mixture model, on the other hand, has parametric flexibility.

There is also a large literature on testing expected utility jointly with stochastic choice, and the impact of these assumptions on parameter interpretation (Loomes & Sugden, 1998; Apesteguia & Ballester, 2018; Webb, 2019). This relates back to the point above: the mixture may reflect the joint assumption of how choice departs from deterministic EV.

1. So then how should we interpret the correlation that the authors report?

On one side we have the impact of the binary/ternary treatment which demonstrates some impact of the low value alternative on a binary choice probability. This may reflect some deep flaws in existing theories of choice, or it may simply reflect some departure from purely deterministic expected value maximization that existing theories can address. We have no theory to connect it to, so we cannot tell. On the other side of the correlation, we have a mixture between additive and multiplicative preferences over risk. This result may reflect two distinct neural processes at work, or it may simply reflect a misspecification of the manner in which humans perceive and aggregate attributes of a lottery (or even just the stimuli in this experiment) by these two extreme candidates (additive vs. EV). Again, this would entail some departure from purely deterministic expected value maximization that existing theories can address.

It is entirely possible that the authors are reporting a result that points to the more exciting of these two possibilities. But it is also possible (and perhaps more likely) that the correlation is more mundane. The paper does not guide us to theories that predict such a correlation, nor reject any existing ones. In my opinion, we should be striving for theoretically-driven analyses of datasets, where the interpretation of results is clearer.

1. Finally, the results from these experiments might not have external validity for two reasons. First, the normative criterion for multi-attribute decision-making differs depending on whether the attributes are lotteries or not (i.e. multiplicative vs additive). Whether it does so for humans is a matter of debate. Therefore if the result is unique to lotteries, it might not be robust for multi-attribute choice more generally. The paper largely glosses over this difference and mixes literature from both domains. Second, the lottery information was presented visually and there is literature suggesting this form of presentation might differ from numerical attributes. Which is more ecologically valid is also a matter of debate.

Minor Issues:

The definition of EV as a normative choice baseline is problematic. The analysis requires that EV is the normative choice model (this is why the HV-LV gap is analyzed and the distractor effect defined in relation to it). But if the binary/ternary interaction effect can be accounted for by curvature of a value function, this should also change the definition of which lottery is HV or LV for that subject!

References

Apesteguia, J. & Ballester, M. Monotone stochastic choice models: The case of risk and time preferences. *Journal of Political Economy* (2018).

Block, H. D. & Marschak, J. Random Orderings and Stochastic Theories of Responses. Cowles Foundation Discussion Papers (1959).

Khaw, M. W., Li, Z. & Woodford, M. Cognitive Imprecision and Small-Stakes Risk Aversion. *Rev. Econ. Stud.* 88, 1979-2013 (2020).

Loomes, G. & Sugden, R. Testing Different Stochastic Specifications of Risky Choice. *Economica* 65, 581-598 (1998).

Luce, R. D. Individual Choice Behaviour. (John Wiley and Sons, Inc., 1959).

Netzer, N., Robson, A. J., Steiner, J. & Kocourek, P. Endogenous Risk Attitudes. SSRN Electron. J. (2022) doi:10.2139/ssrn.4024773.

Robson, A. J. Why would nature give individuals utility functions? Journal of Political Economy 109, 900-914 (2001).

Webb, R. The (Neural) Dynamics of Stochastic Choice. Manage Sci 65, 230-255 (2019).

Reviewer #3 (Public Review):

Summary:

The way an unavailable (distractor) alternative impacts decision quality is of great theoretical importance. Previous work, led by some of the authors of this study, had converged on a nuanced conclusion wherein the distractor can both improve (positive distractor effect) and reduce (negative distractor effect) decision quality, contingent upon the difficulty of the decision problem. In very recent work, Cao and Tsetsos (2022) reanalyzed all relevant previous datasets and showed that once distractor trials are referenced to binary trials (in which the distractor alternative is not shown to participants), distractor effects are absent. Cao and Tsetsos further showed that human participants heavily relied on additive (and not multiplicative) integration of rewards and probabilities.

The present study by Wong et al. puts forward a novel thesis according to which interindividual differences in the way of combining reward attributes underlie the absence of detectable distractor effect at the group level. They re-analysed the 144 human participants and classified participants into a "multiplicative integration" group and an "additive integration" group based on a model parameter, the "integration coefficient", that interpolates between the multiplicative utility and the additive utility in a mixture model. They report that participants in the "multiplicative" group show a negative distractor effect while participants in the "additive" group show a positive distractor effect. These findings are extensively discussed in relation to the potential underlying neural mechanisms.

Strengths:

- The study is forward-looking, integrating previous findings well, and offering a novel proposal on how different integration strategies can lead to different choice biases.
- The authors did an excellent job of connecting their thesis with previous neural findings. This is a very encompassing perspective that is likely to motivate new studies towards a better understanding of how humans and other animals integrate information in decisions under risk and uncertainty.
- Despite that some aspects of the paper are very technical, methodological details are well explained and the paper is very well written.

Weaknesses:

- The authors quantify the distractor variable as "DV - HV", i.e., the relative distractor variable. Do the conclusions hold when the distractor is quantified in absolute terms (as "DV", see also Cao & Tsetsos, 2023)? Similarly, the authors show in Suppl. Figure 1 that the inclusion of a HV + LV regressor does not alter their conclusions. However, the (HV + LV)*T regressor was not included in this analysis. Does including this interaction term alter the conclusions considering there is a high correlation between (HV + LV)*T and (DV - HV)*T? More generally, it will be valuable if the authors assess and discuss the robustness of their findings across different ways of quantifying the distractor effect.
- The central finding of this study is that participants who integrate reward attributes multiplicatively show a positive distractor effect while participants who integrate additively show a negative distractor effect. This is a very interesting and intriguing observation. However, there is no explanation as to why the integration strategy covaries with the

direction of the distractor effect. It is unlikely that the mixture model generates any distractor effect as it combines two "context-independent" models (additive utility and expected value) and is fit to the binary-choice trials. The authors can verify this point by quantifying the distractor effect in the mixture model. If that is the case, it will be important to highlight that the composite model is not explanatory; and defer a mechanistic explanation of this covariation pattern to future studies.

□ Correction for multiple comparisons (e.g., Bonferroni-Holm) was not applied to the regression results. Is the "negative distractor effect in the Additive Group" (Fig. 5c) still significant after such correction? Although this does not affect the stark difference between the distractor effects in the two groups (Fig. 5a), the classification of the distractor effect in each group is important (i.e., should future modelling work try to capture both a negative and a positive effect in the two integration groups? Or just a null and a positive effect?).